

How Many More Labels Are Needed? Minimax Evaluation under Selective Labels

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The same visible record can hide different risks

Observed accuracy alone cannot distinguish these two completions.

Illustrative cohort	10%-error completion	20%-error completion
Total predictions	1,000	1,000
Visible correct labels	500	500
Pending labels	500	500
Errors hidden among pending	100	200
Completed cohort error fraction	10%	20%

The audit protocol

Outcome-dependent missingness is allowed; a committed protocol creates a usable intervention.

1 · Commit the ledger

Freeze the predictor and cohort. The withholding mechanism publishes a truthful partial record and cannot alter it after queries begin.

2 · Audit pending cases

Use private randomness to query at most k distinct unresolved cases. Outcomes must already exist and be accessible.

3 · Test a risk claim

Control false alarms at δ for one deadline. Power is the probability of detecting the planned higher-risk alternative.

Missingness relative to the risk gap sets the frontier

Risk gap $\varepsilon = p - a$; missing fraction $\beta = b / n$, with $0 < \beta < 1$.

Truthful audit regime	Fixed $k \geq 1$ as n grows	Interpretation
$0 < \beta < \varepsilon$	Power $\rightarrow 1$	The hidden pool cannot absorb the excess risk.
$\beta = \varepsilon$	Power $\rightarrow 1$	One query can resolve a critical ambiguity.
$\beta > \varepsilon$	Exact binomial testing frontier	Extra visible records need not remove hidden ambiguity.

- The critical no-query comparison also needs a $\sqrt{n}$ -scale condition.

A finite design needs exactly 57 queries

1,000 predictions · 500 pending · 10% versus 20% risk · at most 5% false alarms

Budget	Certified statement	95% power target
56 queries	Every adaptive policy: power < 94.901%	Impossible
57 queries	A supplied rule: power ≥ 95.243%	Achievable
56 queries, $n \rightarrow \infty$	Minimum for 95% limiting power	Asymptotic only

- Finite conclusion assumes truthful labels and count-tail risk bounds.

Knowing label quality can save hundreds of queries

Numerically inverted budgets for 95% limiting power; these are not finite-cohort guarantees.

Symmetric error level ϵ	Known common channel	Only conditional bounds
0%	56	56
5%	72	117
10%	94	367
15%	127	5,981
20%	176	No fixed budget escapes the blind region

A risk average is not a risk guarantee

Different assumptions change what the same observations can establish.

Count-tail bounds

Control tails of the total error count. They support the truthful 57-query finite certificate and can preserve native information with noise.

Predictable-risk budgets

Bound cumulative conditional risk. They share the truthful asymptotic frontier but do not inherit the 57-query certificate.

Mean risk only

At the 10% versus 20% design, power is capped at 10%, even with complete labels. More labels cannot repair this assumption gap.

More queries cannot replace a larger cohort

A guarantee at 20% risk does not imply sensitivity to a 12% risk.

Design / information	Alternative risk	Power
Frozen 57-query rule; n = 1,000	20%	95.24%
Same frozen 57-query rule	12%	1.15%
Complete labels; n = 1,000	12%	65.32% optimal
Complete labels; n = 2,645	12%	First cohort size reaching 95%

- A private mixture improves 12%-risk power to 8.66%, while retaining at least 95.017% at 20% risk.

An audit with explicit guarantees

Specify the claim and information assumptions before choosing the budget.

- Fix acceptable risk, planning alternative, power target, and false-alarm level.
- Justify the risk class, pending-count bound, and conditional label reliability.
- Use a matching finite certificate; treat asymptotic numbers as planning benchmarks.
- Check off-design power. Nonrejection does not certify acceptable risk.

Three resources determine what an audit can prove

Count labels, but also count what is known about them.

- Query count controls how much hidden information can be acquired.
- Reliability knowledge controls whether returned labels distinguish risk hypotheses.
- Cohort size controls whether the planned risk gap is statistically detectable.
- All guarantees depend on a declared risk class and committed audit protocol.

Source: github.com/ChorokLeeDev/selective-labels-minimax-iclr2027